

IRAKOZE JEAN PAUL

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[LinkedIn](#) | [GitHub](#) | [Portfolio: EDUwise](#)

PROFESSIONAL SUMMARY

Backend and Machine Learning Engineer with dual degrees in Computer & Software Engineering and Software Engineering (ML specialization). Proven ability to design and deploy full-stack intelligent systems, including an AI-powered LMS with ML-based student risk detection, achieving 84% accuracy. Experienced in building scalable REST APIs, ML pipelines, and real-time features. Seeking backend/ML engineering roles to build production-grade software that drives measurable impact.

TECHNICAL SKILLS

Languages: Python, JavaScript, SQL, HTML/CSS, Dart, Shell Scripting

Backend & APIs: FastAPI, REST API design, WebSockets, JWT authentication, Redis caching

Machine Learning: scikit-learn, Random Forest, Collaborative Filtering, Joblib, Data Pipelines, Feature Engineering

Frontend: React, Flutter

Databases & Storage: PostgreSQL 16, Redis, Firebase/Firestore, MinIO (S3-compatible)

DevOps & Infrastructure: Docker, DigitalOcean, Vercel, Git, CI/CD, Linux

AI Integration: Anthropic Claude API (Haiku), prompt engineering, LLM-powered features

PROFESSIONAL EXPERIENCE

Open Source Software Intern, Kigali, Rwanda

Wikimedia Rwanda, Jun 2024 – Dec 2024

- Developed and shipped backend and frontend features for open-source web tools used by the Wikimedia community
- Performed code reviews, bug triage, and testing to improve software reliability across multiple repositories
- Authored technical documentation for codebases, reducing onboarding time for new contributors
- Collaborated with distributed open-source teams using Git-based workflows and issue trackers

PROJECTS

EDUwise — AI-Powered Learning Management System Live: [eduwise-chi-ruby.vercel.app](#)

ALU Capstone Project | React, FastAPI, PostgreSQL, scikit-learn, Redis, Docker 2024 – 2026

- Architected and deployed a full-stack LMS serving secondary schools with 4 user roles (Student, Teacher, Co-admin, Admin) and role-based access control
- Built an ML pipeline using Random Forest classifier on the Cortez & Silva (2008) dataset, achieving **84% accuracy** in at-risk student detection
- Integrated Anthropic Claude Haiku API for AI-powered quiz generation and intelligent tutoring, reducing teacher workload for assessment creation
- Implemented a collaborative filtering recommendation engine to personalize learning content for each student
- Engineered a real-time WebSocket-based live quiz system with gamification features to boost student engagement
- Deployed backend on DigitalOcean with FastAPI, PostgreSQL 16, Redis caching, and MinIO for file storage; frontend on Vercel

TutorConnect — Tutor-Student Matching Platform

University of Rwanda Capstone Project | Full-Stack Web Application 2024 – 2025

- Built a web platform connecting students with tutors, featuring admin approval workflows, session scheduling, and email notifications
- Implemented a location-based tutor search system, enabling students to find nearby tutors with real-time availability
- Developed a review and rating system to ensure tutor quality and build trust within the platform

Book Sales Prediction App

Mobile Application | Flutter, Firebase, Linear Regression 2023

- Developed a cross-platform mobile app predicting book sales using a Linear Regression model integrated with Firebase backend
- Implemented secure user authentication with Firebase Auth and real-time data storage with Firestore

EDUCATION

African Leadership University (ALU) Kigali, Rwanda

BSc Software Engineering — Specialization: Machine Learning, Jan 2023 – May 2026

- Mastercard Foundation Scholar (3-year full grant)

University of Rwanda (UR) Kigali, Rwanda

BSc Computer & Software Engineering — Specialization: Network Engineering, May 2021 – Oct 2025

- Selected among 5,000 national bursary scholars

École Secondaire Kanombe / EFFOTEC Kigali, Rwanda

High School Diploma — MPC Combination (Math, Physics, Computer Science) Jan 2017 – Nov 2019

- Ranked **4th nationally** with perfect scores: Math (A), Physics (A), Computer Science (A), Entrepreneurship (A) — Aggregate: 73/73

CERTIFICATIONS

- **FinTech Certificate** — Elevandi Education & Singapore Management University
- **CISCO Networking & Cybersecurity Certificates** — Year 1, 2 & 3
- **CyberHub Cybersecurity Training Program**
- **Career Essentials in Tech** — Microsoft & LinkedIn
- **More on my Portfolio**

LANGUAGES

Kinyarwanda (Native) | **English** (Fluent) | **French** (Basic)